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Combating plastic pollution- A critical review

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Abstract- Plastic waste is a major cause in polluting earth, ecosystem and health worldwide. Plastic waste leads to land pollution as it accumulates in the form of litter. It comes from industries and also generates during transport and off-loading. Plastic waste pollutes aquatic environment through littering or transport in water systems. After mixing in oceans, surface currents, tides and waves results in dispersion and re-suspension of plastic waste. Atmospheric fallout is also a reason of widespread micro-plastics. Plastic pollution (macro, micro, and nano-plastic waste) alters natural cycles, exerts biological impacts on various important species, and leads to ecological toxicity. The future, recycling and socioeconomics of plastic are very complex issues. Other than plastic production, misbehaviour of consumers in terms of improper disposal after using plastic is a major contributor in production of plastic waste. Various economic and political factors, ignorant behaviour of governments and stakeholders, different opinions of scientists, and under-reported or overlooked polluters are the major culprits. To overcome plastic pollution problem, change in pre and post consumption behaviour is important. Education, awareness and reduction in the plastic demand altogether with better industrial solutions and improved government policies can do better in solving plastic pollution problem. A modern plastics economy should replace the existing plastic economy for sustainable development. A green or eco-friendly polymer like 'green plastic' is also an option. The policymakers, stakeholders, entrepreneurs, NGOs, citizens and researchers must effort to ensure some sustainable solutions to solve the problem of plastic pollution that will fight to plastic waste related existing issues and future threats as well.

Keywords:Plastic waste, plastic pollution, land pollution, green plastic, sustainable solution

INTRODUCTION

Plastic is universal so does the plastic pollution. Plastic has thermo-mechanical properties; it is less expensive, lightweight, water resistant and long-lasting. Plastic products have benefited the world significantly by revolutionizing daily life. Plastics protect food and reduce food waste, made it possible to design lighter vehicles, and insulated electrical cables for greatest competence. Durable packaging offers a safe storage and transport of food items, toiletries items and various health applications. Plastics are important element of tyre rubber, vehicle brake linings and energy, sewerage, and transport sectors as well.

Plastic products are usually produced for one time use, they are with limited recycling ability and thus contribute in the rapidly growing production and usage globally that finally becomes the cause of huge plastic waste accumulation and plastic pollution.[1] Production and uses of various plastic products mainly single use plastics are not sustainable. A 20 time's growth in the production of plastics in last 5 decades has been reported. Approximately 7000 metric tons of plastics without efficient treatment has been land-filled and contributed to plastic pollution.[2,3] It is predicted that 53 million metric tons plastic waste will be

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produced every year by 2030.[1] Plastic waste is not distributed equally around the globe. China, Indonesia, Philippines, Vietnam, and Sri Lanka produce almost 50% of total global mismanaged plastic waste.[4] Though, it is very difficult to quantify the actual amount of plastic waste in the environment and the global estimates of plastic waste also differ as well.

Single use plastics have less capacity for recycling that leads to production of waste and cause plastic pollution. Low degradation or no degradation, unsustainable production, usages and improper disposal enhances plastic pollution. After released into environment plastics travel to land, river and oceans that makes it worst for nature, wildlife and human health.[5] Plastic pollution imposes very significant harmful effects on physical, chemical and biological systems. Plastic can entangle organisms. Plastics can release toxic chemicals in the environment. Various studies have proven that diabetes, obesity, thyroid dysfunction and reproductive impairment have been observed as a result of the ingestion of micro and nano-plastics.[6] Further research to investigate physical and chemical impact of micro- and nano-plastics on human health is necessary.

Diversity and persistence of plastics

Plastic is polluting soil and aquatic ecosystems that finally affect human health via entering in food chain. Plastics are of different types on the basis of origin, production, thermal and mechanical properties, end-use and molecular arrangement. Macro-plastics (size of >25 mm), degrade into mesoplastics- (size >5 mm to 25 mm), micro-plastics (<5 mm), or nanoplastics (<1 mm or 1000 nm) in the environment. PE (high density polyethylene and low-density polyethylene), PS (polystyrene), PVC (polyvinylchloride), PP (polypropylene) and PET (polyethylene terephthalate) are some available plastic types in the market.[7] Plastic products are diverse as well as the plastic pollution. Plastic products are made up in various sizes, types, colour, and their sources are also different. They are made up of a mixture of monomer, oligomer, and additives like Bisphenol A, Phthalates, diisodecyl phthalate, di-*n*-pentyl phthalate (DPENP), di-*n*-hexyl phthalate (DHEXP) and dicyclohexyl phthalate (DCHP).

Plastics directly or indirectly mix up in water bodies, land, and atmosphere via poorly managed plastic wastes, wastewater treatment plants, sewage effluents, landfills and farming actions.[8] Plastics can also combine with sludge, organic fertilizer, or rocks. Rocks formed of the mixing of sediments and liquified plastics are called as "plastiglomerates".[9] Plasticrust is another combination coated by plastic debris. Marine waste pyroplastics is formed of burnt plastics, for example angular plastiglomerates, more disintegrated roundclasts.[10] Plastic waste can carry pathogens that interact with coral reefs and can cause coral death. Heavy metals were reported in pyroplastics collected from south west England beaches.[10] This type of plastic contamination can have three key aspects: serving as vectors for another substance, being absorbed by various other materials, and forming complex mixtures. It is not clear whether they can naturally break down in the environment after formation, or if their properties and toxic effects will modify over time. Few data are available for intricacy of plastic pollution and natural environment on these issues.

Solutions

Plastic pollution is a big challenge and threat for universe. It is affecting our aquatic systems, land, atmosphere and food chains. Therefore, global efforts are required to solve and overcome the great trouble of plastic pollution.

Effective management can be achieved through a five-step approach: prevention, reuse, recycling, recovery, and disposal.[11] It is a better solution in the management of plastics in comparison to reuse and recycling of plastic materials after their use. Waste collection and recycling is a very important aspect of plastics as it is estimated that 58% of plastic waste at present collected for recycling is just because of informal workers and waste pickers.

"A 5-year delay in implementing action would increase plastic pollution by more than 300 million metric tons," adds Lau.[12]

Degradation of plastics

Once released into the environment, plastics undergo natural degradation through processes for instance photo-degradation, hydrolysis, thermal oxidation and bio-degradation.[13] Some bacteria and fungi can also degrade plastics. In the environment, mechanical abrasion can break plastics into smaller particles. Though natural degradation of such highly complex polymers via any of these methods takes very long time (from decades to centuries) and this persistence is the reason of plastic pollution.

Researchers have developed key analytical steps, including sampling using pumps for sample collection[14],

pre-treatment that is primarily digestion with hydrogen peroxide, acids, alkalis, or enzymes for the elimination of organic material and plankton[15], as well as quantification with stereomicroscopy[16] and identification methods using Raman and Fourier Transform Infrared spectroscopy.[17]

These analytical methods are less efficient in the accurate monitoring and there is no standard, effective and efficient analytical methods to solve the problems related to plastic management because of diversity of the sources and verities of plastic products. Therefore, there is a great need for efficient standardized analytical methods to enrich our understanding of plastic pollution.

Removal of plastic waste

Membrane technologies are considered better and effective in the removal of microplastics in the form of polymeric debris.[18] Methods such as rapid gravity sand filtration, dissolved air flotation, and membrane bioreactors are employed for microplastic removal in Finland.[19] Additionally, chemical coagulation can eliminate microplastics from wastewater.

Another environmentally friendly approach is biodegradation.[20] Initially, a microbial biofilm forms on plastispheres, reducing plastic buoyancy and hydrophobicity. This is followed by bio-deterioration, during which the biofilm continues to develop as the plastic breaks down.[21]

The next stage is bio-fragmentation, where enzymes such as amidases, oxidases, laccases, and peroxidases depolymerize the carbon skeleton of microplastics, weakening them. The resulting monomers are then assimilated into microbial biomass. In mineralization, that is a final step, the polymers completely degrade, and production of CO_2, H_2O, and CH_4 take place.

Microbial biofilms can biochemically degrade microplastics and nanoplastics.[22] This microbiome-assisted degradation process is influenced by various environmental factors, including organic matter, solar irradiation and particulate matter.

Certain fungal species, such as *Zalerion maritimum, Agaricus bisporus, Claviceps fusiformis,* and *Schizophyllum commune*, produce hydrophobins that bind to hydrophobic plastic surfaces. This interaction facilitates the biodegradation of microplastics and allows these fungi to utilize plastics as a source of carbon and energy.[23]

Greener alternatives and bio-based plastic substitutes are urgently needed to replace fossil fuel-based plastics. Increased research into biodegradable and biocompatible alternatives to conventional plastics that have a neutral or positive environmental impact, offers a hope. Bioplastics derived from renewable materials, such as polyhydroxyalkanoates (PHA), polylactic acid (PLA) and cellulose are biodegradable. However, polyethylene (PE), produced from biologically sourced ethane, behaves like conventional plastics regarding degradation. Many bioplastics only biodegrade at high temperatures[24] and can emit methane, a greenhouse gas.[25]

Plastic Education in National Curricula

Introducing education about plastic pollution can make a difference. After being aware of risk associated with plastic pollution a changed behaviour can help in better understanding for sustainable and plastic free environment. An importance of 'Responsible science' must be understood to achieve sustainable development goals.

Revising Extended Producer Responsibility (EPR), a policy mechanism aimed at addressing waste management risk, can enhance efforts to prevent and mitigate plastic pollution. EPR can effectively reduce the health, safety, environmental, and social impacts associated with plastic products.[26]

The global Plastics Treaty aims to eliminate plastic pollution. It should establish committees to develop global standards for plastic products and design end-of-life criteria that address contributing sources, hotspots, and pathways related to the use-phase of secondary microplastics. These standards should include bare minimum requirements for the prevention of abrasion and disintegration.

The new plastic economy emphasizes that improved collection and recycling efforts can significantly reduce the quantity of plastic sent to landfills, leading to lesser plastic in the environment. By increasing production from sustainable sources rather than fossil resources and utilizing energy more efficiently, CO_2 emissions can be minimized.

A smart, innovative, and sustainable plastics industry that effectively manages reuse, repair, and recycling can create more jobs while reduction in the emission of greenhouse gases and dependence on imported fossil fuels can also be achieved.

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